

GAMIDOC: SLAY THE ROADMAP

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Project Name	Slay The Roadmap

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CONTEXT

Aim

The project aims to teach the fundamentals of computer programming through a gamified approach based on structured learning paths, integrating contemporary pedagogical methodologies with playful engagement mechanics to facilitate the acquisition of technical-practical skills. With the inclusion of gamification mechanics, we aim to make learning more fun and the study of topics more consistent.

Domain

The scope of application concerns software development issues, with the integration of structured educational roadmaps that ensure alignment with industry standards and the actual needs of the labor market.

Target User

The platform is intended for a heterogeneous user base ranging from absolute beginners (7+ years) to developers approaching a new language, covering both self-learning contexts and institutional training.

Encouraged Behaviors

The system promotes consistent learning behaviors, pursuit of complete conceptual mastery, development of strategic thinking through the optimized construction of thematic decks, and adaptive problem-solving skills to address complex challenges.

Technology

The technical implementation uses the Flutter framework, ensuring cross-platform compatibility across mobile, web, and desktop devices through a unified codebase, with significant benefits in terms of maintainability and portability.

Modality

The main game cycle follows a structured sequence: topic selection, material study, quiz

completion, acquisition of reward cards, deck construction and optimization, boss challenge, reward and unlocking of the new chapter, in an iterative learning process.

Phase	Duration	Activity	Objective
Studio	6-8 min	Reading materials	Knowledge acquisition
Quiz	3-5 min	Verification and learning	Evaluation Comprehension
Deck Building	2-3 min	Reward Selection	Engaging choices
Boss Fight	10-15 min	Practical application	Skills check

GAME BEHAVIOR

Game Rules

The system implements rules based on progressive unlocking with cognitive dependencies between topics, an 80% quiz pass threshold with unlimited but penalized attempts, and a resource management system that includes available energy each turn (3 or + units) and lives (3 initials).

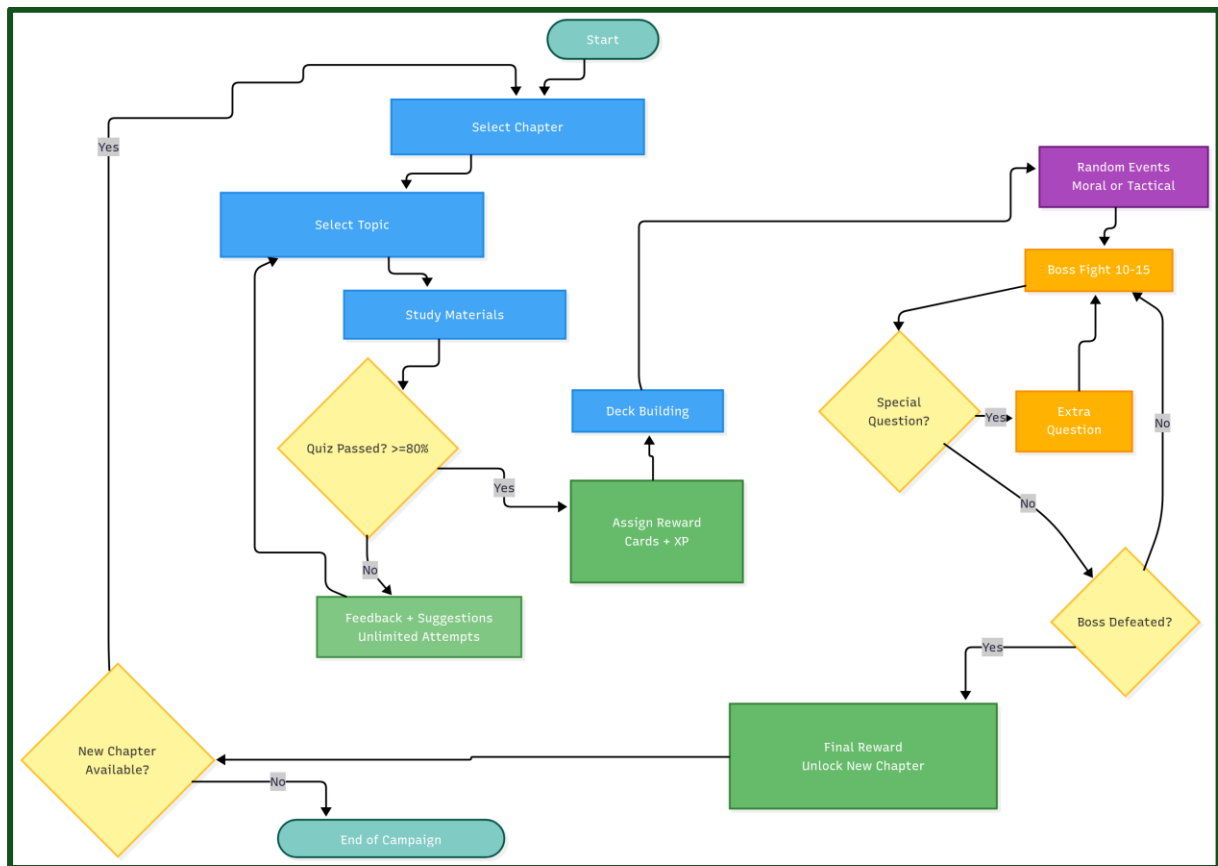
Mechanics

The main mechanics include a card-based system with differentiated types (offensive, defensive, utility), random events, multiple choices, turn-based boss fights that integrate educational questions, and combat phases with special questions for in-depth skill checks.

Progression

The progression is organized into thematic chapters of 5-7 topics each, with a final boss representing the summative test. An experience system (XP) rewards the completion of tasks with level progression and scalar rewards.

Game loop



Short Description of game loop

Select Chapter/Topic: The player's starting point for choosing what to learn.

Study Materials: The core educational content node where the player learns.

Quiz Passed? >= 80%: The first major checkpoint. This decision node determines if the player understood the materials or needs to study again.

Assign Reward Cards + XP: The primary progression node. Passing the quiz gives the player resources (cards, experience) used for the game's challenge/battle elements.

Boss Fight 10-15: The main challenge node. The player uses the rewards earned during the run to overcome a boss fight.

Boss Defeated?: The second major checkpoint. This determines if the player passed the main challenge or

must retry the boss fight.

Final Reward (Unlock New Chapter): The ultimate goal of the loop. Defeating the boss grants access to the next stage of the campaign.

End of Campaign: The final state, reached only when no new chapters are available after a successful boss fight.

GAMIFICATION

Performance

The system has several elements to allow the user to keep track of his progress:

- Level
- Roadmap progress bar
- Game Deck Card Collection

Ecological

The aspects that change the environment around the player are mainly related to his choices. The rewards are alternatives, so a choice is proposed that will outline the continuation of the campaign based on the player's interest. Challenges and quizzes adapt accordingly.

Social

The social dimension is developed through achievement sharing features on social media, in particular it will be possible to show others how you managed to finish a roadmap and with what resources. This will be implemented using deck code exchange systems for collaboration, and community-oriented features for collective learning.

Personal

Personalization takes place through collection systems (cards, badges, achievements), customization of the user profile, and personalized learning analytics that guide the individual training path.

Fictional

The narrative framework incorporates thematic character design for each boss, contextualized educational lore, and a story-driven progression that wraps around the learning experience.

GAME AESTHETICS

The game aesthetic is inspired by minimal dark fantasy. Audio design can include context-differentiated soundtracks and sound effects that provide immediate feedback.

Example Chapter Structure

Chapter	Boss	Domain	Mechanics (opt)
Fundamentals	Gargoyle Gatekeeper	Dart Basics	Base
OOP	Stack Overflow Dragon	Object Programming	Progressive HP growth
Async	Async Demon	Asynchronous programming	debuff time-delayed
UI/UX	Widget Tree Titan	Interface Composition	Multi-layer shield

Card evolution introduces a layer of collection and improvement to the experience. Players can upgrade the rarity of cards through events, following progression from Common through Rare and Epic to Legendary status.

This mechanic encourages both repeated engagement and strategic thinking about which bonus effect to select. The tension between consolidating power and maintaining tactical options generates interesting decisions that persist throughout the entire player experience.

Example Evolution Cards

Rarity	Damage	Bonus Effect
Common	6	Nobody
Rare	10	Nobody
Epic	16	Perforation:5
Legendary	25	Critical Strike: 25%

Dinamycs

Positive dynamics

The *design* encourages players to transcend simple rule execution to develop a true **strategic mindset** and **guided learning**.

Busy players will learn how to optimize their *Deck's* composition not only for brute power, but by strategically adapting it to **each** Boss's *specific weaknesses*, thus turning learning into contextualized problem solving.

A crucial element is the **risk/reward decisions** conveyed by the **textual events** that appear between battles. These narrative moments not only enrich the *lore*, but also present the player with moral or tactical choices that have a direct impact (positive or negative) on resources (energy, cards), requiring careful evaluation of the consequences.

Negative dynamics

Quiz anxiety could lead to avoidance of learning modules. To counter this, the system will provide **contextual suggestions** after a couple of incorrect attempts, requalifying the mistake as a step towards learning, rather than a penalty.

Competitive stress in the *leaderboards* will be managed by making participation in these leaderboards **optional**, while maintaining the emphasis on individual progression.

Personalization

The system adopts a two-pronged personalization model: aesthetic and motivational. Aesthetically, players can customize their **User Profile** and collect **vanity elements** (icons, *frames*, etc.) through unlocked *Badges* and *Achievements*.

On a deeper level, *Personalization manifests* itself in mechanics that intentionally reward different playstyles.

Adaptability

The concept of **Adaptability** ensures that the system is able to adjust challenges and content to **constantly maintain the player's "flow" state** , avoiding frustration or boredom.